

Tyler Lacy

Electrical Engineering Student · NASA Lunabotics Systems Engineering Lead · U.S. Army Veteran

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SUMMARY

Electrical Engineering student, U.S. Army veteran (Secret security clearance), and award-winning NASA Lunabotics systems engineering lead. Combines hands-on electro-mechanical tool reliability, high-pressure fluid testing, and work-order/inventory management experience with C++, Python, and Linux embedded systems development. Proven track record driving campus STEM engagement and leading multidisciplinary teams through systems engineering milestones (PDR/CDR). Seeking a research-focused engineering internship.

EDUCATION

Lone Star College — Associate of Science (Engineering Track)

Expected Spring 2027 · Honors College

Relevant coursework: Programming Fundamentals I & II (C++), General Chemistry I, University Physics I, Calculus II, Macroeconomics

TECHNICAL SKILLS

Programming & Systems	C++, Python, Linux (Ubuntu/Pop!_OS), Git/GitHub, ROS 2, Bash, Embedded C
Hardware & Electronics	ESP32, LoRa (915 MHz), Raspberry Pi, Jetson Orin Nano, Arduino/Teensy, LiDAR, I2C/SPI/UART protocols
Design & Prototyping	Autodesk Fusion 360 (CAD/CAM), OnShape, 3D printing (FDM/PLA+/PETG-CF), manual/CNC machining, calipers, micrometers, GD&T basics
Testing & QA	Hydrostatic & pneumatic pressure testing, seal/thread redress, work order tracking (ERP/SAP), 5S, lean QA/QC

RESEARCH & ENGINEERING PROJECTS

Systems Engineering Team Lead — NASA Lunabotics Rover

2025 – Present

- Led an award-winning multidisciplinary engineering team through NASA Systems Requirement Review (SRR) and Preliminary/Critical Design Reviews (PDR/CDR) for an autonomous regolith excavator.
- Engineered 4-motor drivetrain integration and sensor power distribution using microcontrollers, relay buses, and 2D LiDAR navigation sensors.
- Formulated risk mitigation matrices, budget allocations (\$1,000+), and physical integration testing protocols.

Project Lead — Campus ESP32 Solar Energy & HVAC Optimization Mesh

In progress

- Architected a 10-node off-grid LoRa/ESP32 mesh network across campus to correlate outdoor solar radiation with indoor HVAC electrical draw.
- Programmed low-power C++ firmware with dynamic multi-hop routing, sensor polling, and automated data transmission to an edge gateway to pinpoint campus energy waste.

Fluid System Pressure Testing & Tool Reliability — Weatherford International / Independent

2022 – 2024

- Executed hydrostatic pressure testing protocols up to rated limits on high-pressure fluid tools, ensuring leak-free seal integrity prior to field deployment.

- Logged equipment maintenance, part teardowns, and certification records via digital work order systems; tracked staging inventory across product lines.

Autonomous Drive Developer — VEX U Robotics Platform

2025

- Developed modular C++ drivetrain control algorithms for a 6-wheel tank-style robot chassis.
- Programmed precise 15-second autonomous routines and tuned PID velocity response curves for driver control under match constraints.

PROFESSIONAL EXPERIENCE

R&M Technician III — Weatherford International

Odessa, TX · 2022 – 2024

- Inspected, redressed O-rings/threads, and hydrostatically pressure-tested high-pressure liner-hanger running tools to ensure zero-defect redeployment.
- Generated and completed digital work orders, managed staging inventory, and collaborated with rig operators to reduce non-productive time (NPT).
- Streamlined warehouse parts allocation and improved 5S throughput, building 2–6 fully certified cement heads per day.

Warehouse & IT Manager — Bass Tool & Supply

Houston, TX · 2020 – 2022

- Managed inventory receiving and fulfillment across 300,000+ SKUs while leading a 10-person logistics team.
- Maintained hardware infrastructure, deploying Windows/macOS workstations and executing secure profile migrations.

LEADERSHIP & MILITARY SERVICE

25U Signal Support Systems Specialist — U.S. Army National Guard

Secret security clearance (inactive)

- Operated, deployed, and maintained tactical communications networks, radio systems, wire/fiber networks, and secure encryption hardware.
- Awarded the National Defense Service Medal and Army Service Ribbon.

Event Manager & Lead — ASME Chapter, Lone Star College

2025 – Present

- Manage chapter build budgets, organize campus outreach events, and co-lead Lunabotics robotics initiatives to expand student participation in engineering.

Historian — Computer Science Club, Lone Star College

2025 – Present

- Enhanced LSC–CyFair STEM engagement by organizing hands-on technical workshops, maintaining public code archives, and documenting club engineering initiatives.